

SONAR — STUDENT ONLINE NAVIGATION AND ROUTING

A Real-Time Campus Bus Tracking, ETA Prediction, and Boarding Alert System

A

SYNOPSIS REPORT OF PROJECT

(SONAR — Campus Transportation System)

For the course **FRONT END WEB DEVELOPMENT (26CA205PCB)**

In partial fulfillment of requirements for the degree of

MASTER OF COMPUTER APPLICATIONS (MCA)



SUBMITTED TO: Mahima Tayal

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1. ABSTRACT

SONAR (Student Online Navigation And Routing) is a lightweight, browser-based campus transportation management and tracking application designed to streamline daily commuting for university students. Conventional campus transit relies heavily on printed timetables, informal messaging groups, or manual inquiries. These approaches lack visibility into real-time vehicle movement, unexpected delay conditions, and precise boarding windows.

SONAR resolves these challenges by providing a unified, client-side digital interface. The application models four distinct campus transit corridors—South, North, East, and West Lines—representing stops and vehicle positions via an interactive inline SVG map. A client-side simulation engine calculates bus progress between consecutive nodes, while an integrated ETA engine generates continuous arrival estimates. Furthermore, the system includes a personalized boarding-stop dashboard and sends an automated notification alert whenever a bus reaches a 5-minute proximity threshold to the student's selected stop.

Core Technical Focus: Implemented entirely using modern semantic HTML5, responsive CSS3, and modular ES6 JavaScript without heavy external frameworks or server-side dependencies. Persistence is managed via the browser Local Storage API, allowing offline operational support after initial load.

2. MOTIVATION

The conceptualization of SONAR stems from practical challenges observed in daily institutional transit management, focusing on real-world web utility, user convenience, and lightweight software architecture.

Relevance to Real-World Web Solutions

Modern transit systems leverage dynamic tracking. Students face uncertainty and extended wait times at stops and predictive ETAs. SONAR translates fleet-stop data into an ultra-lightweight client-side application suitable for closed campus routine status inquiries directed to transport staff. SONAR provides personalized stop monitoring management concepts into an ultra-lightweight client- and automated 5-minute alerts, while reducing side web application suitable for closed campus routine status inquiries directed to transport staff. transit environments.

Importance for Students & Office

Alignment with Emerging Web Technologies

Demonstrates serverless client-side architecture by performing vector math, simulation routines, and state management natively inside the browser engine. Uses browser Local Storage for zero-friction client-side data persistence.

3. INTRODUCTION

Campus transport serves as a critical support service for academic institutions, transporting hundreds of students daily across various regional routes to a central university campus. SONAR provides an interactive digital solution that transforms static route information into a live dynamic dashboard.

Domain & System Scope

Operating in the *Intelligent Transportation Systems (ITS) & Web Applications* domain, SONAR models 4 primary routes converging at KIET campus, covering 17 unique boarding stops across 23 node points.

Objectives

- Primary Objectives:
 - To build an interactive SVG vector map representing campus routes, stops, and dynamic vehicle locations.
 - To engineer a client-side ETA calculation engine that dynamically evaluates remaining time based on route progression and delay variables.
 - To provide a personalized dashboard focused on the student's designated boarding stop with an automated 5minute proximity alert.
- Secondary & Technical Objectives:
 - To deliver route-wise timetables enriched with operational status tags (e.g., *On-time*, *Late*, *En route*, *Arrived*, *At depot*).
 - To implement client-side user registration, login, and preference persistence via Local Storage without external backend overhead.
 - To support simulation speed controls (1\times, 2\times, 4\times) and pause/resume capabilities for testing and operational simulation.

Findings & Key Insights

Initial analytical findings demonstrate that complex telemetry tracking can be effectively simulated in client-side software using linear interpolation algorithms. This eliminates expensive GPS hardware requirements during academic prototyping while delivering high fidelity transit UI/UX.

Future Scope

Future extensions include integrating live hardware GPS telemetry feeds, transitioning into a Progressive Web App (PWA) with push notifications, introducing machine-learning models for traffic-aware ETA adjustments, and developing an administrative analytics console for fleet dispatching.

4. ANALYSIS PHASE

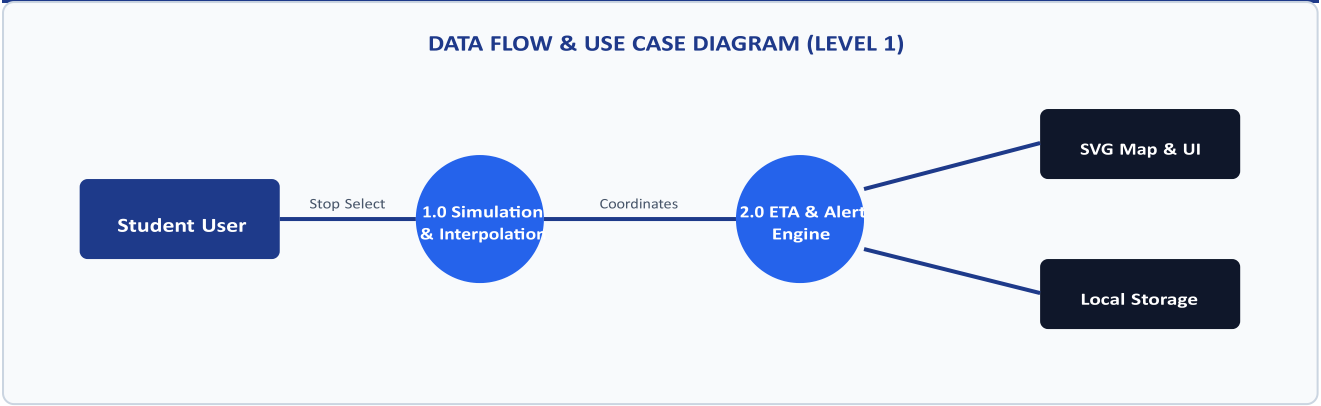
The analysis phase defines the functional capabilities, system constraints, and data flows required to establish a reliable client-side transport tracking application.

Requirement Analysis

Category	Detailed Requirement Description
Functional	Multi-route SVG map rendering, real-time position interpolation, stop-specific ETA computation, user registration/login, saved boarding stop preference, and automated 5-minute proximity alert triggers.
Non-Functional	Initial page load under 1.0 second, fully responsive across mobile, tablet, and desktop viewports, zero external framework dependencies, and offline operation via browser Local Storage.

Data Flow Diagram (DFD) & System Architecture

The system transforms static route datasets and user inputs into real-time visual UI updates:



System Flow

User Registration/Login → Boarding Stop Selection → Simulation Execution → Interpolated Coordinate Calculation → ETA Computation → 5-Min Proximity Check → UI Boarding Alert Trigger.

5. TECHNOLOGY STACK (WEB DEVELOPMENT)

Layer	Technology	Purpose & Application Role
Frontend Markup	HTML5 (Semantic)	Provides structured document hierarchy for views (Home, Track, Timetable, Profile).
Styling	CSS3 (Flexbox/Grid)	Responsive layout design, status badge styling, and multidevice viewports.
Logic & Engine	Vanilla JavaScript (ES6+)	Powers position interpolation logic, ETA calculation routines, and state handlers.
Graphics	Inline SVG	Renders vector bus routes, node stops, and moving vehicle location markers.
Persistence	Web Local Storage API	Persists user profiles, account credentials, and saved boarding stop preferences locally.
Deployment	Static Server / GitHub Pages	Enables serverless execution and instant client-side execution.

6. SYSTEM DESIGN & METHODOLOGY

SONAR follows an **Iterative Web Development Lifecycle** consisting of 6 defined stages:

- Requirement Analysis & Route Modeling:** Mapping campus routes, stop node sequences, and timing baselines.
- UI/UX Layout Design:** Structuring responsive grid systems and modular dashboard component views.
- SVG Vectorization:** Drafting scale vector paths for South, North, East, and West transit corridors.
- Simulation Logic Implementation:** Coding linear interpolation algorithms for vehicle movement and ETA routines.
- Persistence & State Integration:** Connecting Local Storage API for profile management and stop preference saving.
- Validation & Cross-Device Testing:** Testing speed toggles (1×, 2×, 4×), alert thresholds, and mobile viewports.

7. EXPECTED OUTCOMES

- A fully functional, responsive client-side web application for campus transportation monitoring.
- Interactive SVG map rendering dynamic bus progression without server refresh cycles.
- Personalized student dashboard featuring automated 5-minute proximity boarding notifications.
- Robust offline operational capability using client-side storage mechanisms.

8. REFERENCES

1. World Wide Web Consortium (W3C), *HTML5 & SVG 2.0 Specification Standards*, 2024.
2. Mozilla Developer Network (MDN Web Docs), *Client-side Storage & Web APIs Guide*.
3. Sommerville, I., *Software Engineering*, Pearson Education, 10th Edition.
4. KIET Group of Institutions, *Institutional Transport Routes & Schedule Records*.